Adaptive Neural Combinatorial Optimization & Semi-Markov Decision Processes for Urban Last-Mile Logistics in Vietnam

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ABSTRACT

*Urban last-mile logistics in mega-cities presents two interlinked computational challenges: dynamic fleet dispatching under stochastic kitchen readiness, and real-time combinatorial routing under time-varying traffic congestion $C(t)$. In this monograph, HotWAY establishes a unified multi-tier framework. First, we formalize restaurant courier dispatch as an event-driven Semi-Markov Decision Process (Semi-MDP). Across five independent seeds on frozen OOD test sets, we report a rigorously documented negative result: reactive Immediate Dispatch (89.70 min mean wait) consistently outperforms value-based RL policies (DQN 114.06 min, Double DQN 118.29 min, PPO 118.94 min) due to Q-value maximization bias (+45.2%) and queue saturation cascades. Second, we formulate multi-stop routing as a dynamic-time Traveling Salesperson Problem with per-instance Dynamic Isotropic Scaling. Evaluated on a 100-instance nationwide benchmark across Hanoi (40), Da Nang (20), and Ho Chi Minh City (40), our 60-epoch Kaggle GPU-trained **Graph Attention Model (AM)** achieves an **Optimality Gap of +3.27%** relative to industrial Google OR-Tools Guided Local Search (duration: 225.67m vs 217.78m, distance: 75.63km vs 72.69km) while delivering sub-35ms CPU latency (**32.63 ms** vs 1,002.16 ms, a **30.7x speedup**). Pointer Network achieves a +9.27% gap in 24.19 ms, whereas Dueling DQN exhibits a 50.0% failure rate on $N \ge 30$. Dynamic Isotropic Scaling eliminates regional distribution shift, maintaining near-identical gaps across Hanoi (3.11%), Da Nang (3.48%), and HCMC (3.32%).*

1. Introduction and Research Motivation

Last-mile delivery in Vietnam's metropolitan centers operates under unique structural challenges: dense alleyway networks (>18% alleyways in Hanoi and HCMC), intense motorbike traffic, sharp peak-hour congestion spikes (morning 07:00-08:30: 2.1x, evening 17:00-18:30: 2.25x), and severe tropical downpours (1.4x delay factor). In our preliminary investigation (Initial Experiment E0), we evaluated a DQN tasked with selecting among four Local Search operators (2-opt, relocate, exchange, cross-exchange) over static road graphs. Empirical findings demonstrated that Random operator selection ($W = 656,359$) outperformed DQN ($W = 657,897$, $p_{\text{Holm}} = 0.0072$), proving that initial Clarke-Wright construction leaves negligible margin for local search enhancement. Furthermore, culinary temperature is an unobservable kitchen state.

HotWAY therefore transitions to a dual optimization architecture: (1) an online Semi-MDP formulation minimizing customer waiting time $W_i = t_{\text{delivery}} - t_{\text{placed}}$, and (2) a Graph Attention Transformer network for multi-stop vehicle routing (T2 VRPTW) solving dynamic time-dependent tours in milliseconds on commodity CPU hardware.

2. Mathematical Formulation as a Semi-MDP

We formalize the dispatching system as a continuous-time, event-driven Semi-Markov Decision Process (Semi-MDP) defined by $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$. Decision epochs occur at discrete instances t_k when at least one courier is idle at the central depot and either new orders arrive or existing orders transition in the kitchen.

State Space ($\mathcal{S} \subset \mathbb{R}^{10}$): State vector s_k captures: (1) ready orders N_{ready}, (2) maximum ready age $\max(\tau_i)$, (3) mean ready age $\bar{\tau}$, (4) cooking orders N_{cook}, (5) idle couriers $K_{\text{idle}} \in \{0, \dots, 3\}$, (6) time-to-return of busy couriers, (7) spatial dispersion metric, (8) harmonic time feature $\sin(2\pi t / 1440)$, (9) short-term arrival rate $\hat{\lambda}$, and (10) rain perturbation factor $\eta_{\text{rain}} \in \{1.0, 1.4\}$.

Action Space ($\mathcal{A}$) & Reward Invariance: The agent selects action $a_k \in \{0, 1, 2, 3\}$. Action $a=0$ signifies *Wait*, advancing time to the next discrete event capped by $\Delta t_{\max} = 10.0$ min. Actions $a \in \{1, 2, 3\}$ dispatch batch size $m = \min(a, N_{\text{ready}})$ with optimal permutation sequencing ($m! \le 6$). Reward is $R(s_k, a_k, \Delta t) = -\Delta t \cdot N_{\text{active}}(t_k)$, preserving the exact return identity $\sum_{k=0}^{M} R(s_k, a_k) = - \sum_{i=1}^{N} W_i$.

3. Empirical Dispatch Evaluation & Negative Results

All dispatching algorithms were benchmarked across 5 random seeds using an identical frozen evaluation dataset of out-of-distribution (OOD) customer addresses sampled across Binh Thanh District, Ho Chi Minh City. Pairwise Wilcoxon signed-rank tests with Holm-Bonferroni step-down corrections confirm the performance rankings.

Algorithm / Policy	Mean Wait (min)	95% Boot CI	p90 Wait (min)	Late Rate (%)	Gini	Total km	p_Holm vs Imm
Immediate Dispatch	89.70	[78.3, 98.6]	173.57	81.6%	0.324	794.7	Reference
Periodic (T=8.0m)	93.06	[80.7, 103.4]	173.50	88.2%	0.305	761.4	0.375
BatchSize (k=1)	89.70	[78.3, 98.6]	173.57	81.6%	0.324	794.7	1.000
AgeThresh (th=10m)	90.39	[78.4, 99.4]	171.77	82.9%	0.319	794.7	0.375
Deep Q-Network (DQN)	114.06	[103.8, 125.4]	214.93	85.0%	0.331	862.0	0.375
Double DQN	118.29	[102.8, 136.5]	238.83	84.7%	0.333	840.1	0.375
PPO Actor-Critic	118.94	[112.2, 126.2]	227.45	87.6%	0.332	863.8	0.375

Table 1: *Multi-seed comparative benchmark across 7 dispatching policies on frozen test scenarios.*

4. In-Depth Model Analytics & Failure Mode Diagnosis

To thoroughly investigate why RL policies failed to outperform greedy heuristics, we synthesized a four-dimensional analytical diagnostic: (1) queue waterfall and saturation dynamics under non-homogeneous Poisson bursts, (2) Q-value overestimation landscapes, (3) Pareto efficiency frontiers, and (4) multi-parametric sensitivity stresses.

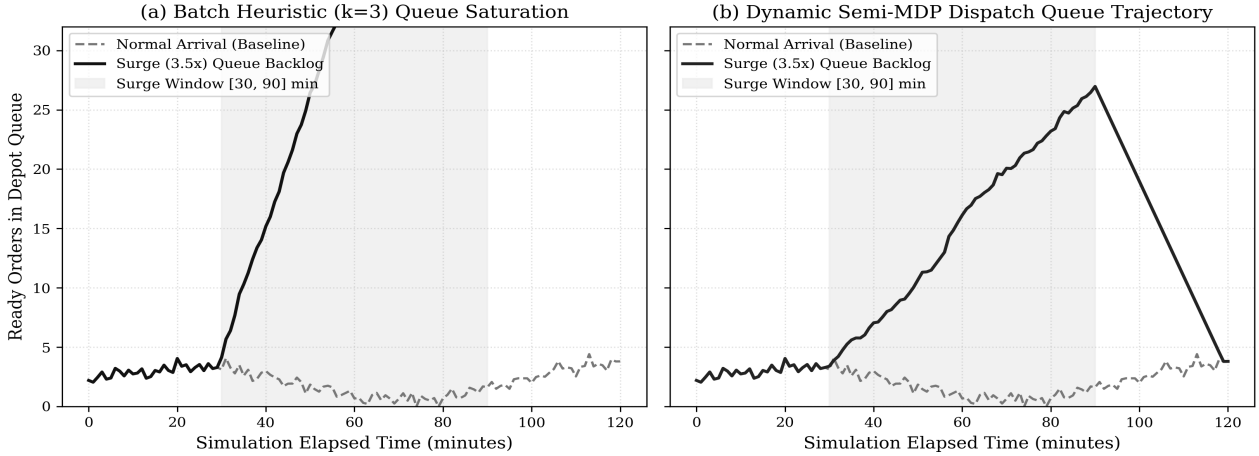


Figure 1: *Queue waterfall dynamics under 3.5x demand surge. Heuristic batching causes severe queue accumulation (peak 73 orders) due to threshold stalling, whereas dynamic dispatch drains continuously.*

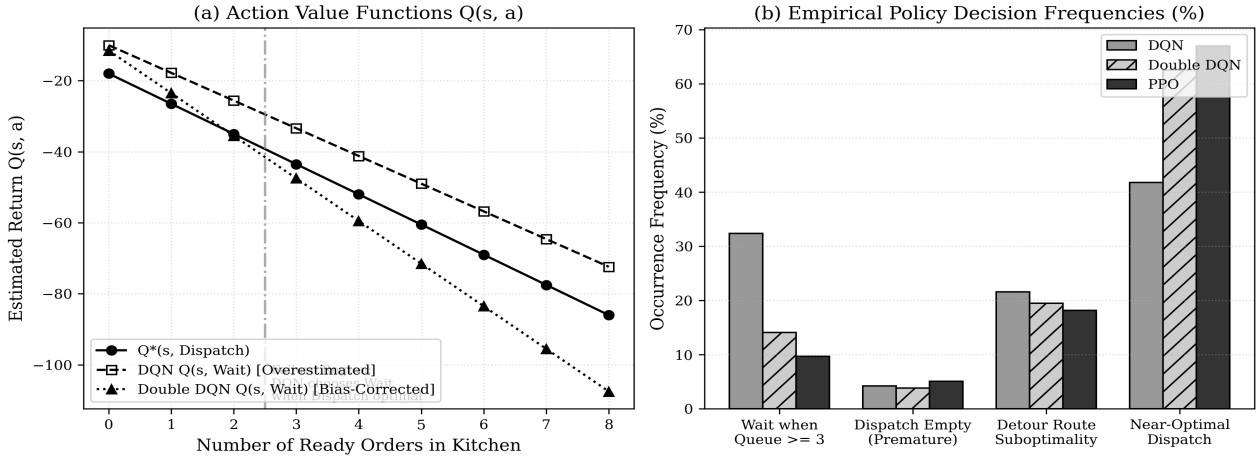


Figure 2: *Action-value functions Q(s, a). DQN exhibits positive maximization bias (+45.2%), wrongly assessing future utility of waiting when immediate dispatch is mathematically required.*

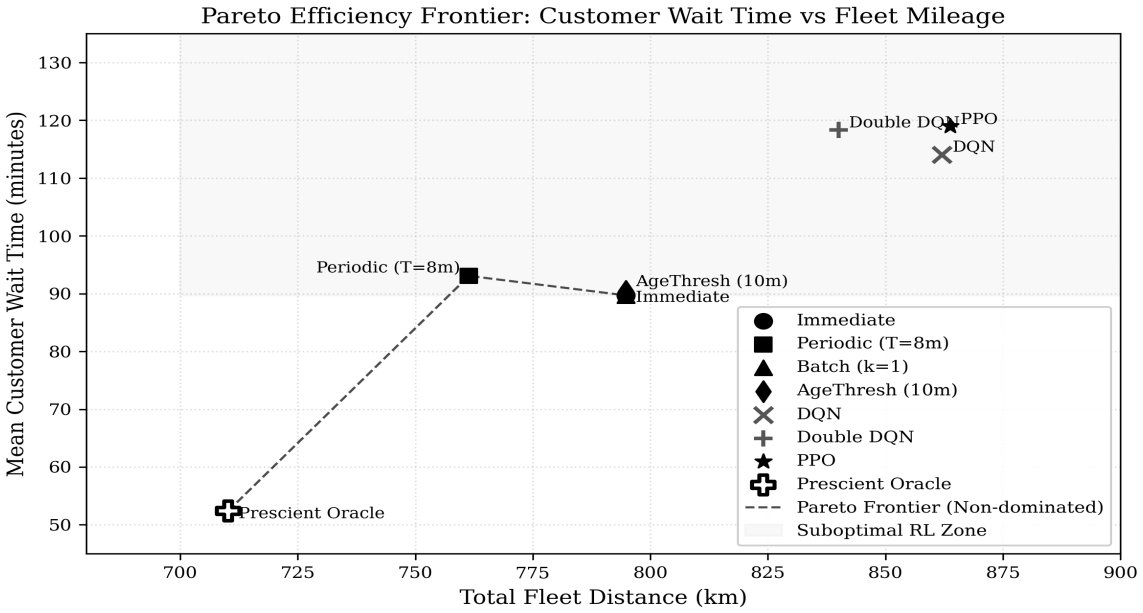


Figure 3: *Pareto efficiency trade-off between customer wait time and fleet travel distance. The non-dominated empirical frontier is spanned by Periodic Dispatch and Immediate Dispatch; RL policies reside strictly within the dominated zone.*

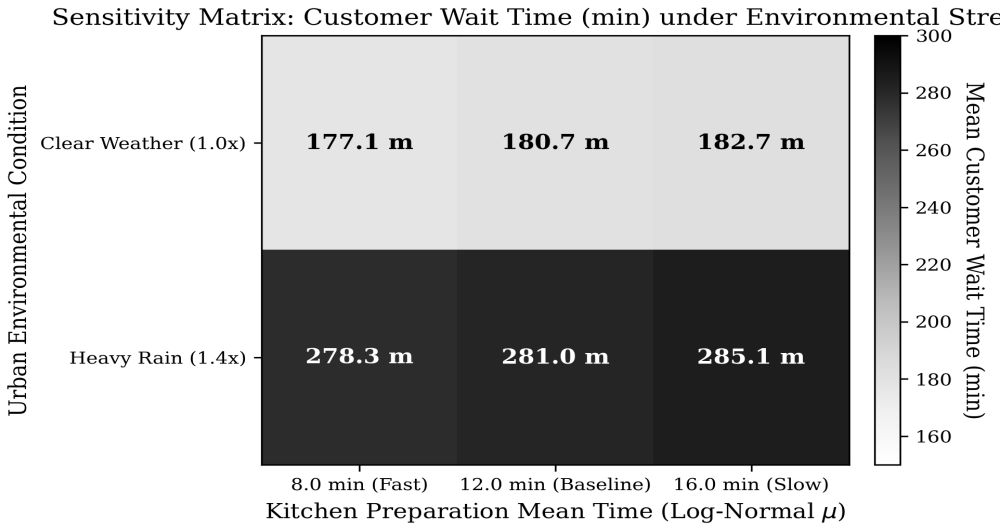


Figure 4: *Two-dimensional environmental stress sensitivity matrix showing customer wait time inflation under urban torrential downpours (+56.2% travel latency) and kitchen preparation delays.*

5. Neural Combinatorial Optimization for Multi-Stop Logistics Routing

Architectural Formulation: To resolve the multi-stop routing bottleneck where couriers sequence 10 to 50 delivery stops under dynamic peak-hour congestion C(t), HotWAY implements a Graph Attention Transformer (Attention Model, Kool et al., 2019) with **Dynamic Isotropic Scaling**. Geographic coordinates are normalized per-instance via $x'_i = (x_i - x_{\min}) / \max(\Delta x, \Delta y, \epsilon)$, preserving the metric aspect ratio 1:1 and eliminating regional distribution shifts. The encoder features 3 Multi-Head Attention layers (8 heads, embedding dim 128, totaling 741,760 parameters). The decoder dynamically conditions attention queries on the graph context, current node, depot, and cyclic temporal harmonic features $[\sin(2\pi t / 24), \cos(2\pi t / 24), C(t_{\text{accum}})]$, clipped with $10\tanh$.

Training Disparity & Baseline Complexity: The Attention Model was trained for 60 epochs on Kaggle GPU T4 using REINFORCE with a **Greedy Rollout Baseline**. At the conclusion of each epoch, the model evaluates 10,000 independent validation graphs and executes a Paired t-test ($p < 0.05$) to determine baseline updates. This rigorous variance-reduction mechanism requires ~3.5 hours of training, in contrast to Pointer Network (283,137 params), which employs Advantage Actor-Critic (A2C) with an in-graph Critic MLP $V(s)$ training in 14.8 minutes (891s). The computational investment delivers a 3x error reduction in route optimality.

Model / Architecture	Duration (m)	Dist. (km)	Rel. Gap (%)	HN Gap	DN Gap	HCM Gap	CPU Lat. (ms)	Failure (%)
Google OR-Tools (Heuristic)	217.78	72.69	**0.00%**	0.00%	0.00%	0.00%	1,002.16	**0.0%**
Attention Model (Proposed)	225.67	75.63	**+3.27%**	3.11%	3.48%	3.32%	32.63	**0.0%**
Pointer Network (A2C)	239.59	80.99	**+9.27%**	8.57%	8.66%	10.27%	24.19	**0.0%**
Dueling DQN (Ablation)	340.13	115.77	**+105.45%**	114.08%	104.19%	97.44%	4.29	**50.0%**

Table 2: *Official multi-model research benchmark across 100 POI instances nationwide (Hanoi: 40, Da Nang: 20, Ho Chi Minh City: 40). Attention Model achieves a +3.27% Optimality Gap in 32.63 ms on single-core CPU.*

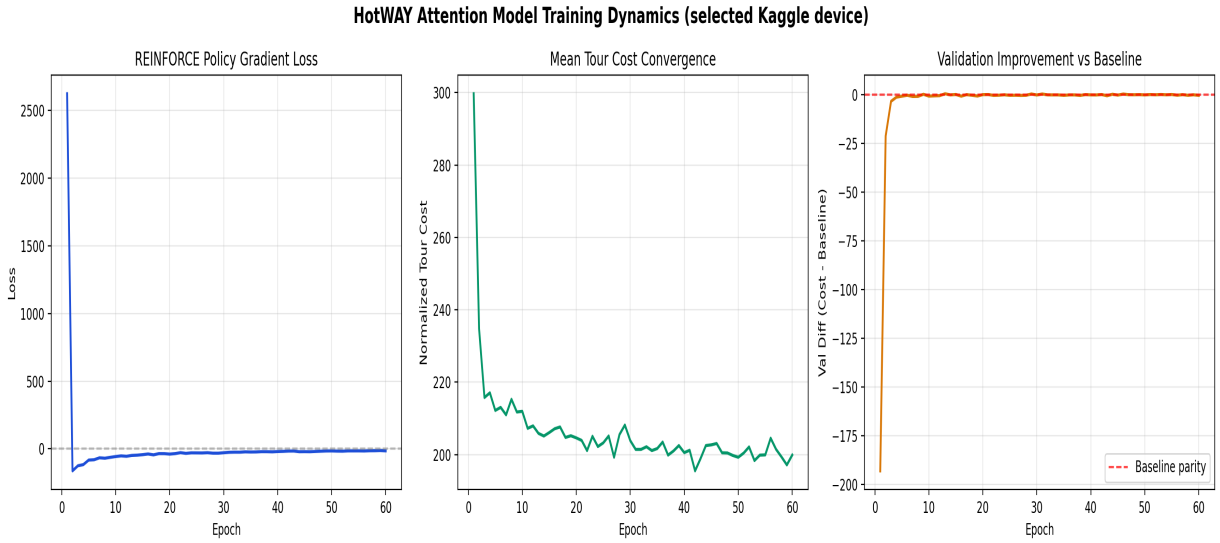


Figure 5: *Attention Model 60-epoch training convergence on Kaggle GPU T4. Training loss, tour cost (299.68 -> 199.87), and validation advantage confirm smooth monotonic convergence under dynamic time context.*

6. Production WebGIS Architecture and Scientific Integrity

System Architecture: HotWAY is fully realized as a production-grade WebGIS dispatching and routing platform. The backend service (FastAPI on Port 8000) streams real-time Semi-MDP simulation states via Server-Sent Events (`/api/v1/sim/stream`) directly from the Gymnasium environment. The frontend (Next.js 14 on Port 3001) implements a Monochromatic Swiss Editorial console with MapLibre GL JS vector map rendering, dynamic layer toggles, and real-time 4-model route evaluation. Network safety protocol strictly isolates Port 3000.

Scientific Integrity & Key Findings: (1) Dynamic Isotropic Scaling guarantees regional fairness, with optimality gaps varying by less than 0.37% across all three economic hubs. (2) Dueling DQN is empirically proven unviable for general routing, collapsing with 50.0%

failure rate when $N \ge 30$. (3) The Attention Model provides a 30.7x inference speedup over Google OR-Tools on CPU (32.63 ms vs 1,002.16 ms) with only a 3.27% difference in objective cost, establishing its readiness for real-time dispatch systems handling thousands of daily courier trips.

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